Defect Report

BUG-0353 — Synchronization Failure in Authentication: SyncError: conflict detecte...

1. Defect Summary

Defect ID	BUG-0353
Title	Synchronization Failure in Authentication: SyncError: conflict detected on entit...
Severity	Minor
Priority	Medium
Status	VERIFIED
Component	Authentication
Application	APP-007 — InternetBrowser
Reported Date	2023-09-28
Resolved Date	2024-08-06
Assigned To	ENG-068 — Xavier Xavier

2. Traceability

Release	REL-006 — 5.2.7
Test Run	RUN-005 — Regression Cycle 5 — Build 5.1.6
Requirement	REQ-0380 — Compatibility — Dark Mode
Req Priority	High
Test Case ID	TC-1332
Test Case Title	Confirm that the system recovers gracefully after Authentication Failu...
Execution ID	EXEC-02665
Failure ID	FAIL-0521
Failure Type	Synchronization Failure
Environment	ENV-003 — Hub Lab
Build Version	5.3.6

3. Problem Description

During execution EXEC-02665, test case TC-1332 failed with Synchronization Failure. The system exhibited synchronization failure when the server returns a 503 Service Unavailable response.

4. Reproduction Steps

1. Re-connect the phone and confirm auto-recovery.
2. Launch the WearableHealth application on the test device.
3. Check server logs for event code 'EVT-1208'.
4. Simulate network drop using the network emulator profile 'Offline'.
5. Simulate high CPU load using the network emulator profile 'Offline'.

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5. Expected vs Actual Behaviour

Expected	The system should complete playlist shuffle within the defined SLA.
Actual	The system exhibited synchronization failure when the server returns a 503 Service Unavailable respo...

6. Root Cause

Retry logic does not apply exponential back-off; storm of reconnection attempts saturates the gateway.

7. Resolution

Implemented exponential back-off with jitter in the retry middleware.

8. Related Historical Failures

HIST-0012	Synchronization Failure	4.3.6	2021-08-05	15	Open
HIST-0047	Synchronization Failure	6.4.9	2023-03-16	9	Wont Fix
HIST-0050	Synchronization Failure	3.1.9	2024-02-17	7	Wont Fix

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

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